

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Physics

Branch:- CSE

Semester – 1

Sections-CSE/AI/DS/CPS/CS

Mid-Term Examination October – 2025

Subject Name/Code: - Engineering Physics/(CSE/CSE-AI/DS/CPS/CS-1002)

Max. Marks: 20
Time Allotted: 90 min

NOTE: 1. All questions are compulsory.
2. Mark answers number clearly.

S. No.	Question	Marks
1.	(a) Show that, in Newton's rings observed in reflected light, the diameters of the dark rings are proportional to the square root of the natural numbers (i.e., $D_m \propto \sqrt{m}$ where $m=1,2,3,\dots$).	3
	(b) Monochromatic light of wavelength 600 nm illuminates a double slit separated by 0.3mm. Estimate the largest bright fringe that can possibly appear on a screen. <i>Note:</i> Consider interference only. Ignore any effects due to the finite width of the slits.	2
2.	(a) Show that $\frac{1}{1 - \frac{v^2}{c^2}} = 1 + \frac{p^2}{m^2 c^2}$ where v is the velocity of a particle, p its relativistic momentum, m its rest mass, and c the speed of light.	2.5
	(b) A thin transparent film of refractive index $n=1.33$ is coated on a glass plate of refractive index $n=1.52$. The film is illuminated by monochromatic light of wavelength 600 nm at normal incidence from air of refractive index $n=1$. What is the condition for constructive interference in the reflected light?	2.5
3.	(a) There are two identical pendulums: one inside a spaceship flying past the earth at three-fourths the speed of light, and one on earth inside Mr. Anil's laboratory. The pendulums are swinging. Answer the following questions: (i) An astronaut performing an experiment inside the spaceship notices that one complete swing of his pendulum takes 1.80 seconds. According	1.5

		to Mr. Anil, how long does it take for one complete swing of the astronaut's pendulum? (ii) Instead, if each swing in the astronaut's pendulum takes 1.80 seconds as measured by Mr. Anil, how long will it take as measured by the astronaut?	1.5
	(b)	Now, if Mr. Anil looks at his pendulum and observes that it takes 1.80 seconds for one complete swing, how long will it take as measured by the astronaut?	2
4.	(a)	Imagine a universe where the speed of light is $c=65$ km/h and your place of living is 30 km away from IIIT Bhopal. Assume you travel to IIIT Bhopal daily at a typical driving speed of 60 km/h. If you drove at that speed for the round trip to and from IIIT Bhopal, how much would your wristwatch lag your bedroom clock each day?	2.5
	(b)	Let us assume that you travel every day in a Suzuki Dzire car which is roughly 4 m long. If you were travelling at the typical speed mentioned above, how long would your car be, when viewed from outside?	2.5